

Homework- Quest set 2

1. Suppose we roll a fair die. Let X be upface of the roll. Find the pmf and CDF.
2. Let X represent difference between the number of heads and number of tails when a coin is tossed 3 times. What are the probabilities of values that X take?
3. The distribution function of the random variable X is given

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{x}{2} & 0 \leq x < 1 \\ \frac{2}{3} & 1 \leq x < 2 \\ \frac{11}{12} & 2 \leq x < 3 \\ 1 & 3 \leq x \end{cases}$$

- (a) Plot this distribution function.
 - (b) What is $P\{X > \frac{1}{2}\}$?
 - (c) What is $P\{2 < X \leq 4\}$?
 - (d) What is $P\{X < 3\}$?
 - (e) What is $P\{X = 1\}$?
4. Suppose random variable X has the pmf

$$p(x) = \begin{cases} cx & x = 1, 2, \dots, 10 \\ 0 & elsewhere \end{cases}$$

Find c .

5. For each of the following, find the constant c so that $p(x)$ satisfies the condition of being a pmf of one random variable X .
 - (a) $p(x) = c \left(\frac{2}{3}\right)^x, x = 1, 2, 3, \dots$, zero elsewhere.
 - (b) $p(x) = cx, x = 1, 2, 3, 4, 5, 6$, zero elsewhere.
6. Let $p(x) = x/15, x = 1, 2, 3, 4, 5$, zero elsewhere, be the pmf of X . Find $P(X = 1 \text{ or } 2)$, $P\left(\frac{1}{2} < X < \frac{5}{2}\right)$, and $P(1 \leq X \leq 2)$.
7. Let $p(x)$ be the pmf of a random variable X . Find the cdf $F(x)$ of X and sketch its graph along with that of $p(x)$ if:
 - (a) $p(x) = 1, x = 0$, zero elsewhere.
 - (b) $p(x) = \frac{1}{3}, x = -1, 0, 1$, zero elsewhere.
 - (c) $p(x) = x/15, x = 1, 2, 3, 4, 5$, zero elsewhere.
8. Let us select five cards at random and without replacement from an ordinary deck of playing cards.
 - (a) Find the pmf of X , the number of hearts in the five cards.
 - (b) Determine $P(X \leq 1)$.